

Lab program-34:

34. Consider a file system where the records of the file are stored one after another both physically and logically. A record of the file can only be accessed by reading all the previous records. Design a C program to simulate the file allocation strategy.

Code:

```
#include <stdio.h>
#include <stdlib.h>

struct Record {
    int recordNumber;
    char data[256];
};

int main(void) {
    FILE *file;
    struct Record record;
    int recordNumber;

    file = fopen("sequential_file.dat", "wb");
    if (file == NULL) {
        perror("Error opening the file");
        return 1;
    }

    printf("Enter records (Enter '0' as record number to exit):\n");
    while (1) {
        printf("Record Number: ");
        scanf("%d", &record.recordNumber);
        if (record.recordNumber == 0)
            break;

        printf("Data: ");
        scanf(" %255[^\n]", record.data);
        fwrite(&record, sizeof(record), 1, file);
    }
}
```

```

    }
    fclose(file);

    file = fopen("sequential_file.dat", "rb");
    if (file == NULL) {
        perror("Error opening the file");
        return 1;
    }

    while (1) {
        int found = 0;
        printf("Enter the record number to read (0 to exit): ");
        scanf("%d", &recordNumber);
        if (recordNumber == 0)
            break;

        rewind(file);
        while (fread(&record, sizeof(record), 1, file) == 1) {
            printf("Record Number: %d\n", record.recordNumber);
            printf("Data: %s\n", record.data);
            if (record.recordNumber == recordNumber) {
                found = 1;
                break;
            }
        }
        if (!found)
            printf("Record not found.\n");
    }

    fclose(file);
    return 0;
}

```

OUTPUT:

Enter records (Enter '0' as record number to exit):

Record Number: 389

Data: JASWANTH

Record Number: 0

Enter the record number to read (0 to exit): 389

Record Number: 389

Data: JASWANTH

Enter the record number to read (0 to exit): 0

Process exited after 29.88 seconds with return value 0

Press any key to continue . . . |